EDS 223 Homework 2

Exploring patterns of environmental justice

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Table of contents

Part 1: Legacy of redlining in current environmental (in)justice **1**
Load Packages and Source Functions 1
Import Data 2

Part 2: Legacy of redlining in pedstrian and bike accidents **5**
Load data 5
Data wrangle of percent bike or pedestrian crashes in each HOLC region 6
Visualize data 8

Part 1: Legacy of redlining in current environmental (in)justice

Load Packages and Source Functions

```
library(sf)
library(tmap)
library(tidyverse)
library(spData)
library(stars)
library(here)
library(bit64)

#Be sure to run these
#install.packages("sf")
#install.packages("stars")
#install.packages("tmap")
```

Import Data

```
#Import HOLC data as LA_Inequality
LA_Inequality <- sf::st_read(here::here("data", "mapping-inequality","mapping-inequality-los
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/123lookatme/Documents/EDS223-HW2/data/mapping-inequality/mapping-
inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
#Import ej screen for LA inequalities
ejscreen <- sf::st_read(here::here("data", "ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNM
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/123lookatme/Documents/EDS223-HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_C
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
california <- ejscreen |>
  dplyr::filter(ST_ABBREV == "CA")

los_angeles <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))
```

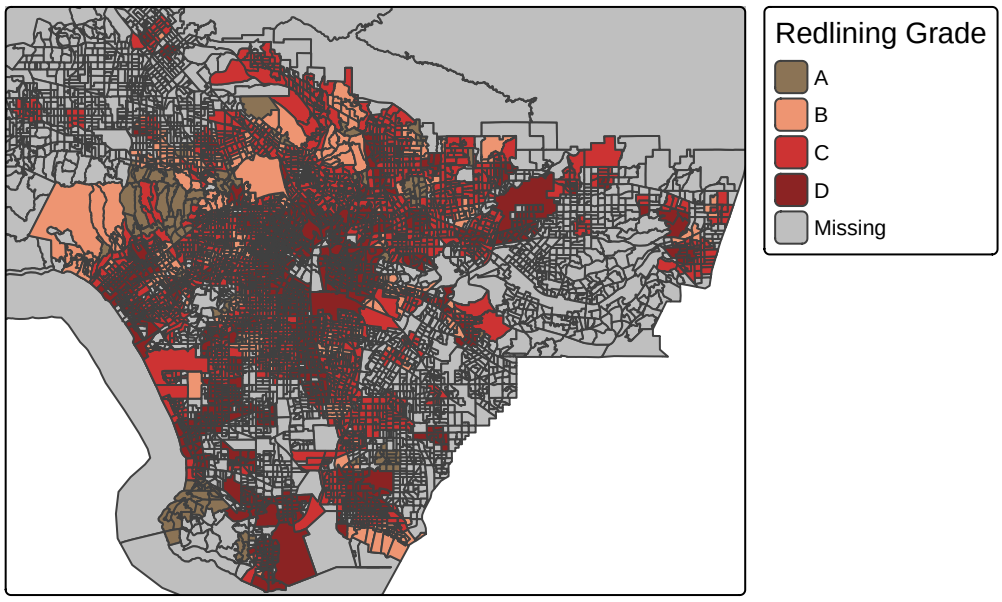
Data wrangle LA inequalities and HOLC grade map

```
#Transform LA_Inequality to same crs as los_angeles
Inequality_transform <- st_transform(LA_Inequality, crs = st_crs(los_angeles))

#Combine LA_Inequality with los_angeles
Combined_LA <- st_join(los_angeles, Inequality_transform, left = TRUE)
```

Map of historical redlining neighborhoods

```
tm_shape(Combined_LA,bbox = LA_Inequality) +
    tm_polygons(fill="grade",
    fill.scale = tm_scale(values = c("burlywood4", "lightsalmon2", "brown3","brown4","honeyde
    fill.legend = tm_legend(title = "Redlining Grade")
    )
```



Create percentage summary table of each HOLC block

```
#Replace na values with NoGrade
Combined_LA <- Combined_LA |> replace_na(list(grade = "NoGrade"))

#Create table
PercentTable <- Combined_LA |>
  count(grade) |>
  st_drop_geometry()

PercentTable <- PercentTable |>
  mutate(n = (100*(n/sum(PercentTable$n)))) |>
  rename(Percent = n)

PercentTable
```

```
    grade  Percent
1       A  4.99555
2       B 13.78505
3       C 34.02314
4       D 14.97552
5 NoGrade 32.22074
```

Low percent income and Percentile Particulate Matter 2.5 within HOLC grades

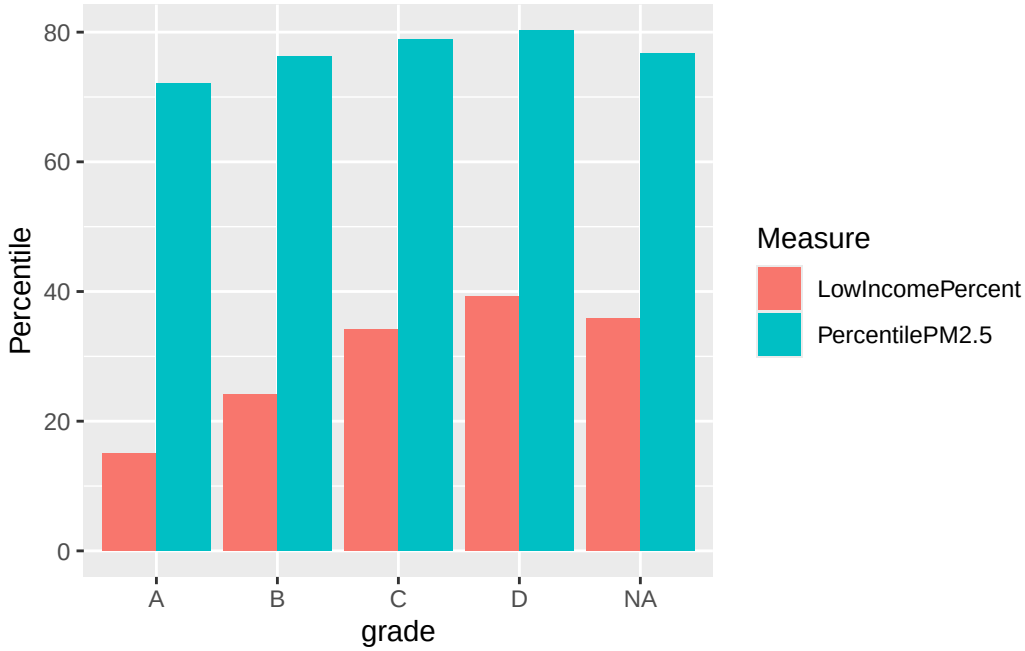
```
OnlyHOLC <- st_join(Inequality_transform, los_angeles, left = TRUE)

LI_and_PM_table <- OnlyHOLC |>
  group_by(grade) |>
  summarize(LowIncomePercent = mean(LOWINCPCT),
            PercentilePM2.5 = mean(P_PM25)) |>
  st_drop_geometry()

LI_and_PM_table <- LI_and_PM_table |>
  mutate(LowIncomePercent = LowIncomePercent*100)

LI_and_PM_table <- LI_and_PM_table |>
  pivot_longer(!grade, names_to = "Measure", values_to = "Percentile")

ggplot(LI_and_PM_table, aes(fill=Measure, y=Percentile, x=grade)) +
    geom_bar(position="dodge", stat="identity")
```



- Result reflection -

Towards the southeast, there seems to higher percentile of particulate matter 2.5, which seems to be generally correlated with a higher redlining grade. Similarly, percent low income also seems to increase as redlining grade goes up. The redlining of these areas may be causational of their low income and particulate matter pollution status.

Part 2: Legacy of redlining in pedstrian and bike accidents

Load data

```
Crashes_raw <- st_read(here::here("data", "SWITRS_LA","Crashes.csv"))
```

```
Reading layer `Crashes' from data source
  `/Users/123lookatme/Documents/EDS223-HW2/data/SWITRS_LA/Crashes.csv'
  using driver `CSV'
```

Data wrangle of percent bike or pedestrian crashes in each HOLC region

```
#Filter pedestrian or bicycle accidents
Crashes_peds_bikes <- Crashes_raw |>
  filter(PEDESTRIAN_ACCIDENT == "Y" | BICYCLE_ACCIDENT == "Y") |>
  filter(POINT_X != 'NA')

# Which columsn have more NA's
sum((Crashes_raw$POINT_X == 'NA'))
```

```
[1] 9484
```

```
sum((Crashes_raw$LATITUDE == 'NA'))
```

```
[1] 97350
```

```
# We'll use xy

#Convert X and Y columns to points, using WGS 84, or code 4326
Crash_pb_trans <- st_as_sf(Crashes_peds_bikes, coords = c("POINT_X", "POINT_Y"), crs = 4326)

#Check CRS is correct
st_crs(Crash_pb_trans)
```

```
Coordinate Reference System:
  User input: EPSG:4326
  wkt:
GEOGCRS["WGS 84",
    ENSEMBLE["World Geodetic System 1984 ensemble",
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        MEMBER["World Geodetic System 1984 (G2296)"],
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]],
```

```
        ENSEMBLEACCURACY[2.0]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    USAGE[
        SCOPE["Horizontal component of 3D system."],
        AREA["World."],
        BBOX[-90,-180,90,180]],
    ID["EPSG",4326]]
```

```
#Transform crash dots to HOLC map CRS
st_crs(Crash_pb_trans) == st_crs(OnlyHOLC)
```

```
[1] FALSE
```

```
Crash_trans_correct <- st_transform(Crash_pb_trans, crs = st_crs(OnlyHOLC))

# Subset points only within HOLC regions
HOLC_Crashes <- Crash_trans_correct[OnlyHOLC, ]

#Left join
Crashes_left_join <- st_join(HOLC_Crashes, OnlyHOLC) |>
  select(id = ID, region = grade)

# Summarise HOLC grades with most crashes
Crashes_Table <- Crashes_left_join |>
  filter(!is.na(region)) |>
  group_by(region) |>
  summarise(n_points = n()) |>
  st_drop_geometry()

Crashes_percent <- Crashes_Table |>
  mutate(Percentage = 100*(n_points/sum(Crashes_Table$n_points)))
Crashes_percent$region <- as.factor(Crashes_percent$region)
Crashes_percent$Percentage <- as.double(Crashes_percent$Percentage)
```

```
Crashes_percent
```

```
# A tibble: 4 x 3
  region n_points Percentage
  <fct>     <int>      <dbl>
1 A          4973      0.791
2 B         28788      4.58
3 C        393933     62.6
4 D        201283     32.0
```

Visualize data

```
ggplot(Crashes_percent, aes(x = region, y = Percentage, fill = region)) +
  geom_bar(stat = "identity") +
  labs(x = "HOLC region", y = "% of bike and pedestrian crashes", title = "Bike and Pedestria
  theme_minimal()
```

